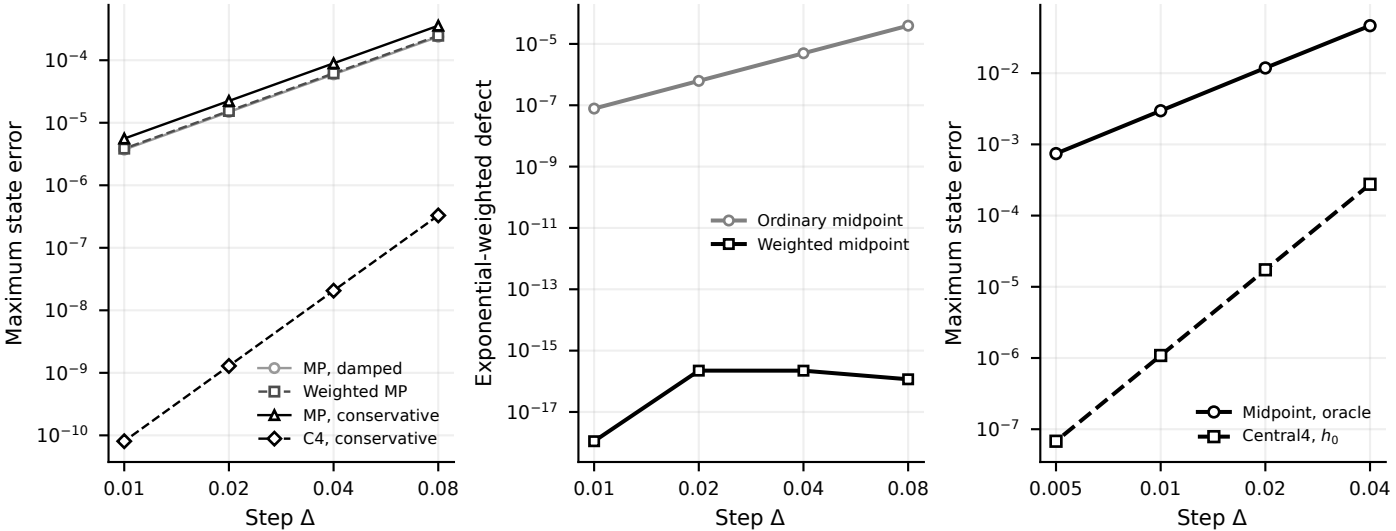
(a) Matrix check, $n = 2$
Maximum state error
10^{-4}
10^{-5}
10^{-6}
10^{-7}
10^{-8}
10^{-9}
10^{-10}
MP, damped
Weighted MP
MP, conservative
C4, conservative
0.01
0.02
0.04
0.08
Step Δ
(b) Weighted structure
Exponential-weighted defect
10^{-5}
10^{-7}
10^{-9}
10^{-11}
10^{-13}
10^{-15}
10^{-17}
Ordinary midpoint
Weighted midpoint
0.01
0.02
0.04
0.08
Step Δ
(c) Nonlinear magnetic h_0
Maximum state error
10^{-2}
10^{-3}
10^{-4}
10^{-5}
10^{-6}
10^{-7}
Midpoint, oracle
Central4, h_0
0.005
0.01
0.02
0.04
Step Δ